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ETE 3351

Lab 6 and extra credit

Advanced Fourier Spectrum Lab: FFT analyzer, Specialized Waveforms (Pulse, Sinc, Cardiac, Nois), filter effects related to Fourier Analysis

Introduction

This lab covers looking at signals in two different ways. The first way is from the time domain -voltage vs. time, a normal oscilloscope view. The other way is from the frequency domain- frequency in Hz on the X-axis and magnitude on the y axis, allowing us to observe what frequencies make up that signal. We can see based off of the frequency domain that any waveform can be built by adding sinusoids of different frequencies and amplitudes. FFT (fast fourier transform) is a mathematical tool that converts the time signal into frequency content (so we can view the different frequencies making up a signal) using the fourier transform

Also covered in this lab is sampling, as to ensure accurate frequency analysis a proper sampling rate is required. Scopes take many samples of the signal, and it is critical to consider the sampling rate. Essentially, more samples means greater accuracy of representing the situation. Recall that according to the nyquist theorem, to avoid getting an alias, we need $f_{\text{sampling}} \geq 2f$ to accurately represent the frequency and $f_{\text{sampling}} \geq 10f$ to accurately represent the shape (but a greater frequency of sampling is ideal).

This lab also examines noise signals and how noise contains power across a wide range of frequencies. The effects of filtering the waveform with a low pass filter are seen, and we can see that progressively reducing the filter cut off frequency means that

higher frequency harmonics are removed, causing pulse waveforms to lose harmonics and start to approach a sinusoidal shape (as a pure sinusoid does not have any harmonics)

Lab Setup and Measurements

1. As with any new instrument or measurement the first thing you want to do is check a known signal to see if its output or display matches what you expect. For our test case we will use a sine wave so we can view it both on the scope in the time domain and its frequency content with the FFT function. Since you are familiar with the Spectrum Analyzer the FFT Analyzer is not a complex thing but to make it easier a step by step tutorial is included at the end of this document so you should go through that first starting with a 10kHz 1Volt-RMS Sinewave. Please follow the FFT STEP by STEP tutorial at the end of this document

PART 1: FFT Tutorial: Rigol DG5071 ARB Generator and DS2202A (do this first)

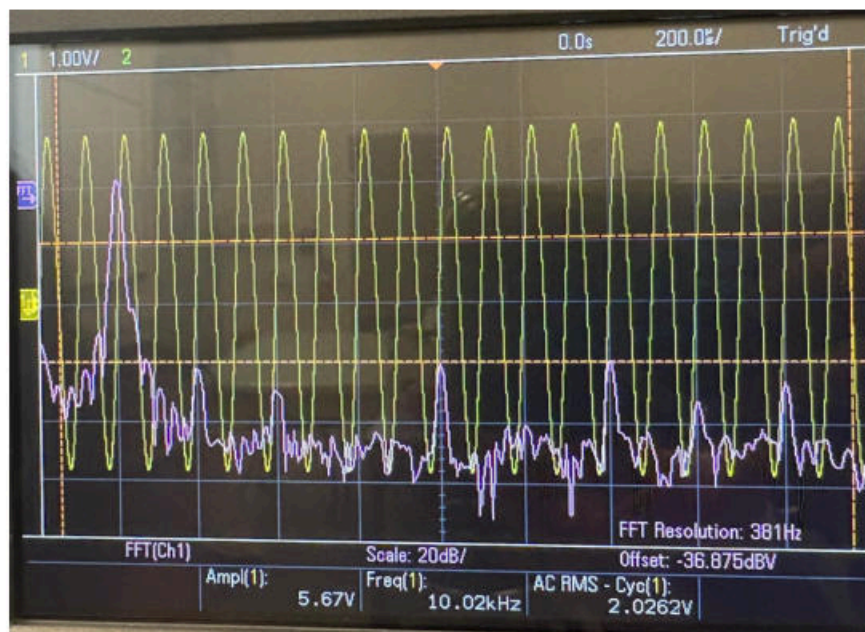
Set the signal input for a 1 Volt RMS Sinewave at 10kHz display signal waveform on scope. Increase the scope time base so that at least 10 cycles of the wave are shown on screen, this is an important step as to few cycles, like 1, will provide a very poor display. Turn on the FFT function/button, the purple display is a magnitude vs. frequency display similar to the spectrum analyzer, although it is difficult to see our sinusoid.



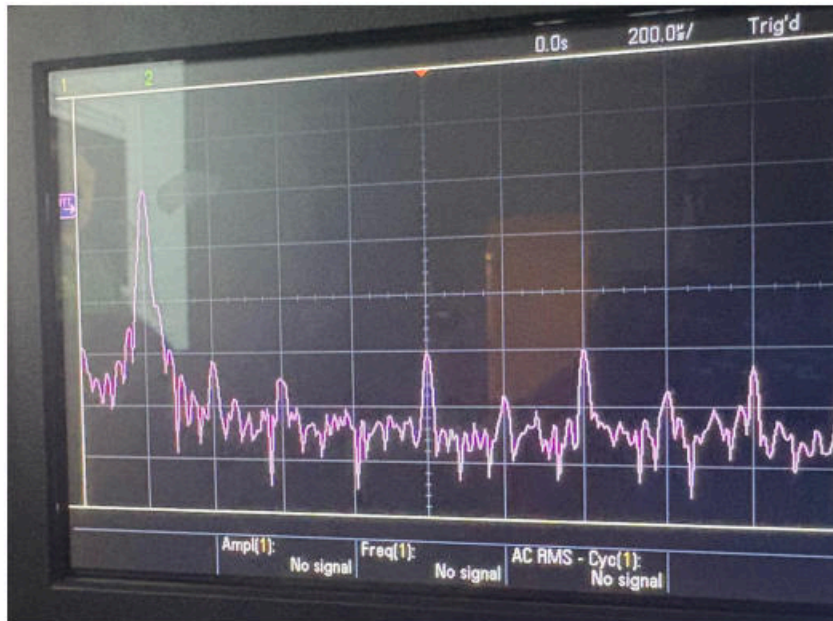
IMG 1. Expected results



IMG 2. *Expected results*



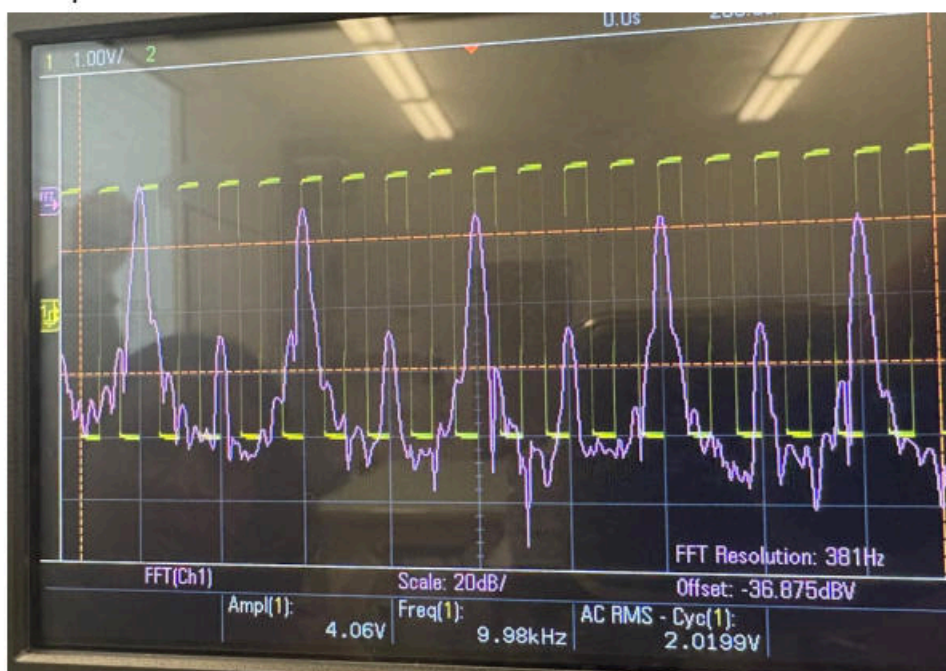
IMG 3. *Results- 10kHz, 1vrms on function wave generator*



IMG 4. Results

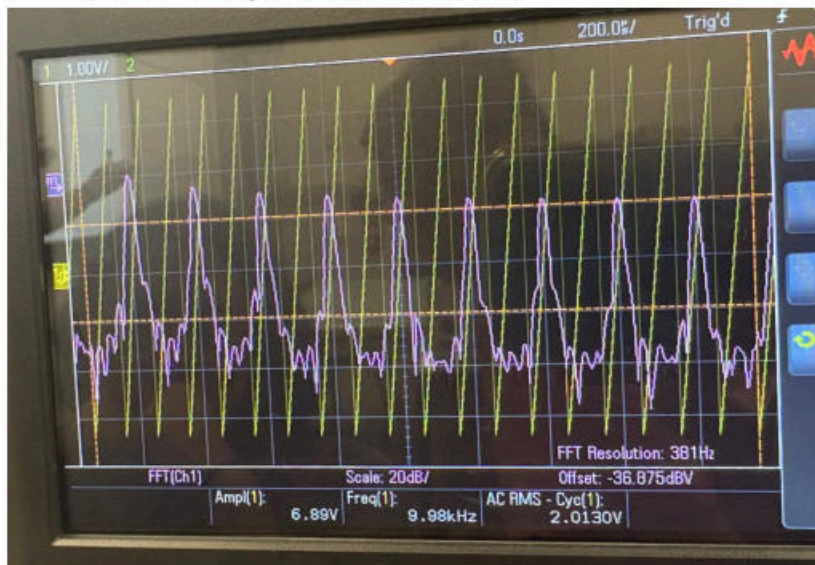
From here you can view any waves like the Triangle wave or pulse waves with different duty cycles

2. Square waves:

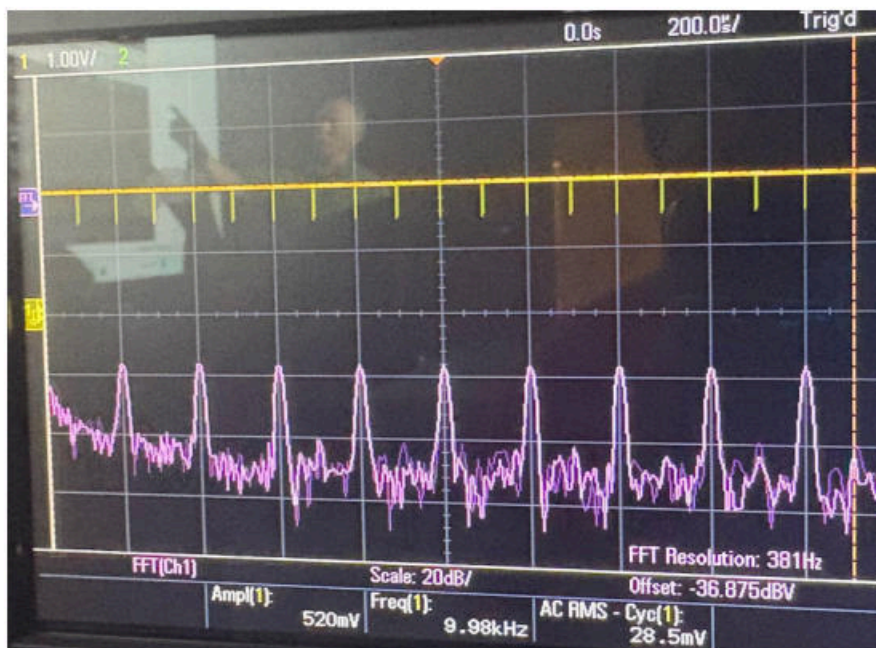


IMG 5. Results

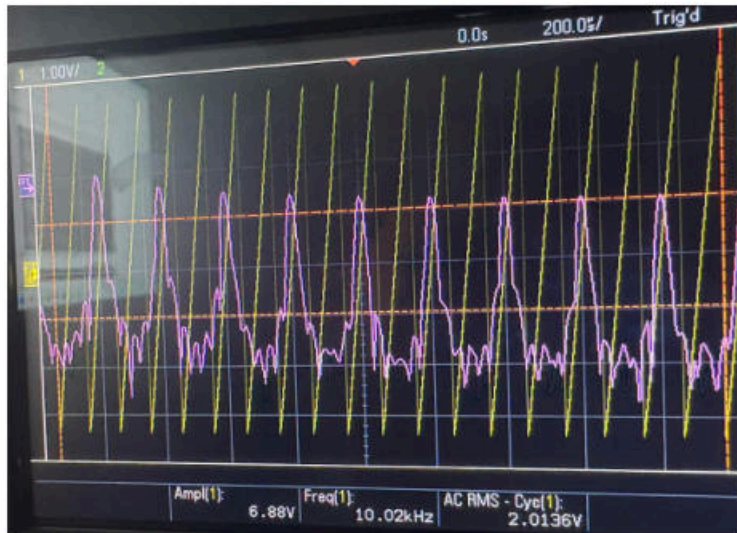
3. For pulse, triangle waves and ramp



IMG 6. Results

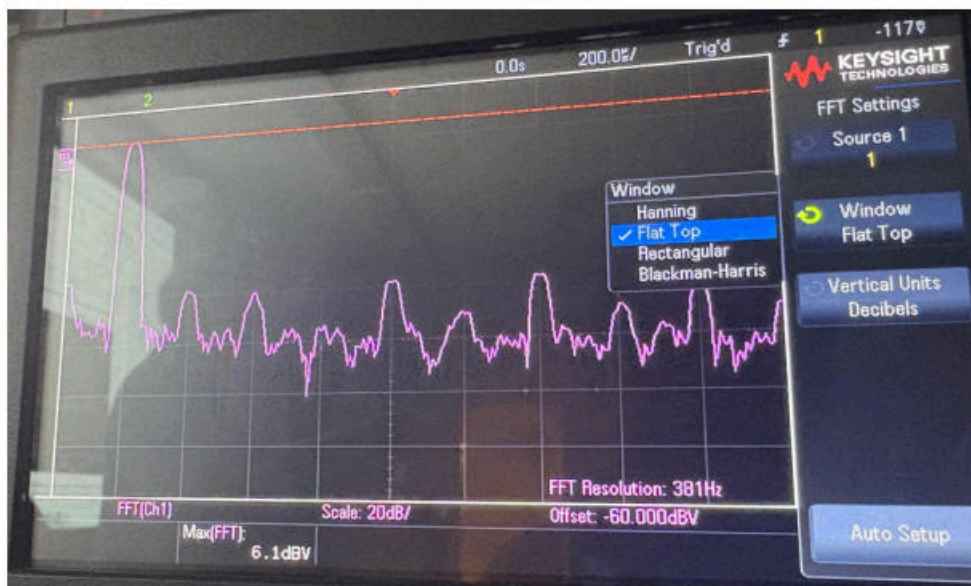


IMG 7. Results

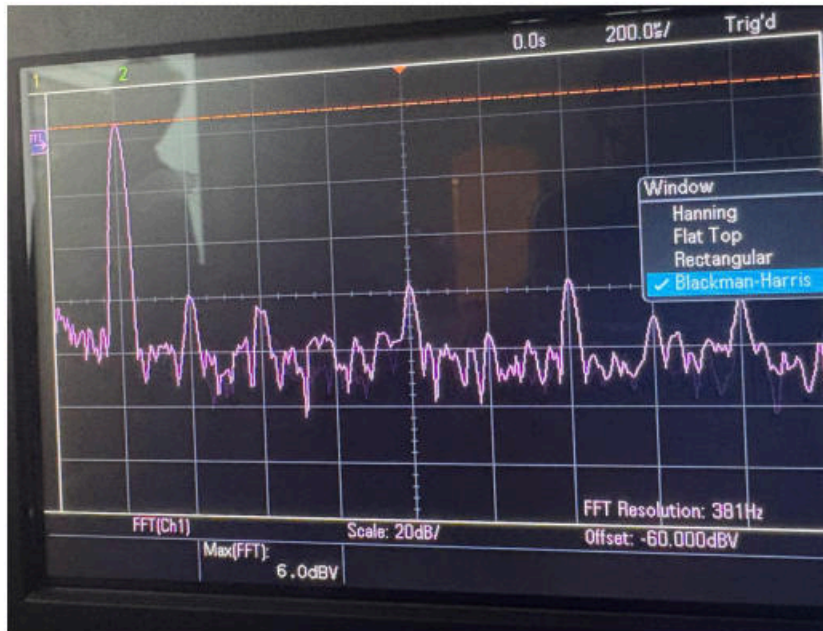


IMG 8. Results

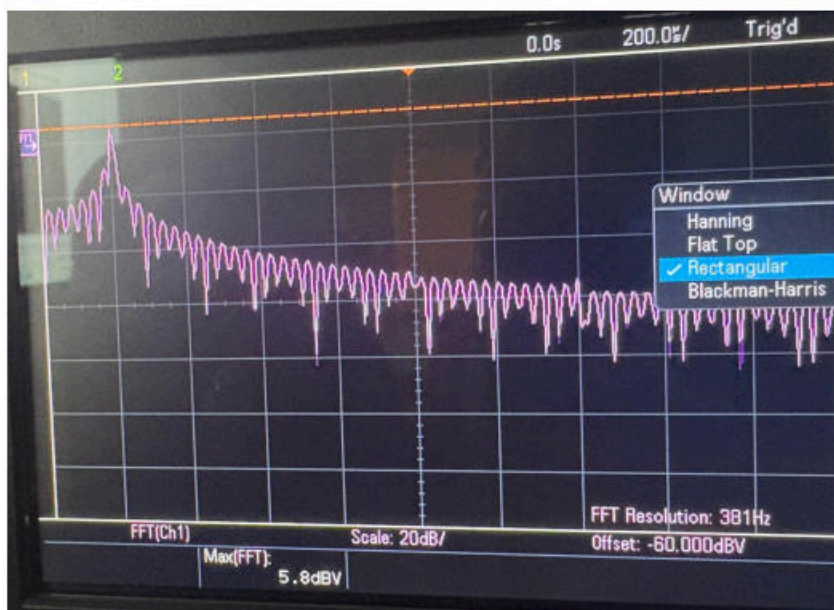
A special soft options under settings are options called Windowing or Rectangular, Hanning, Hamming, and Blackmun all referring to mathematical signal processing windowing techniques (https://en.wikipedia.org/wiki/Window_function), and their authors, each one gives a slightly different display, Blackmun is pretty good for frequency displays but it is your choice which one you select. Finally at the bottom one can change from dB to VRMS display which was similar to the Linear Amplitude display on the SA



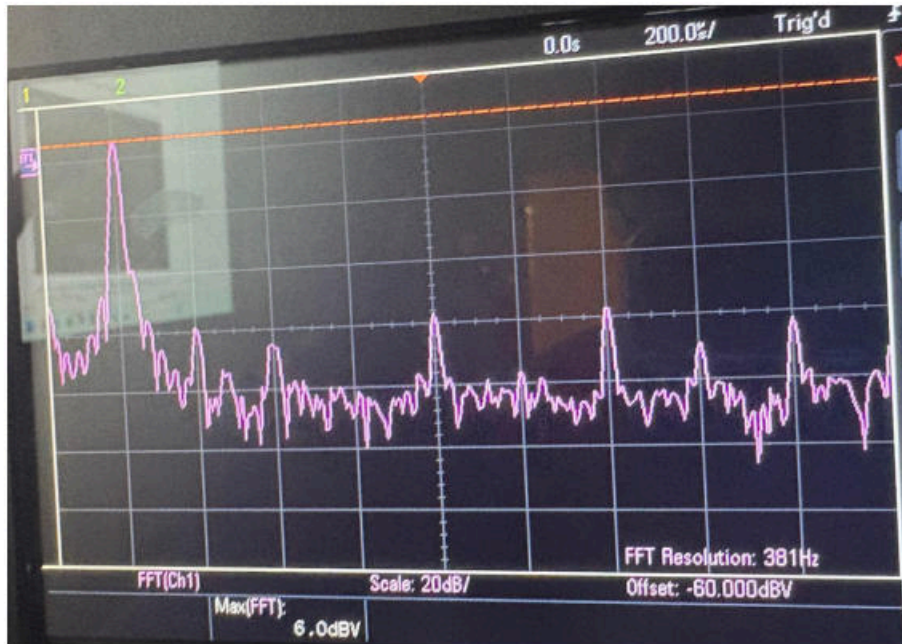
IMG 9. Results



IMG 10. Results

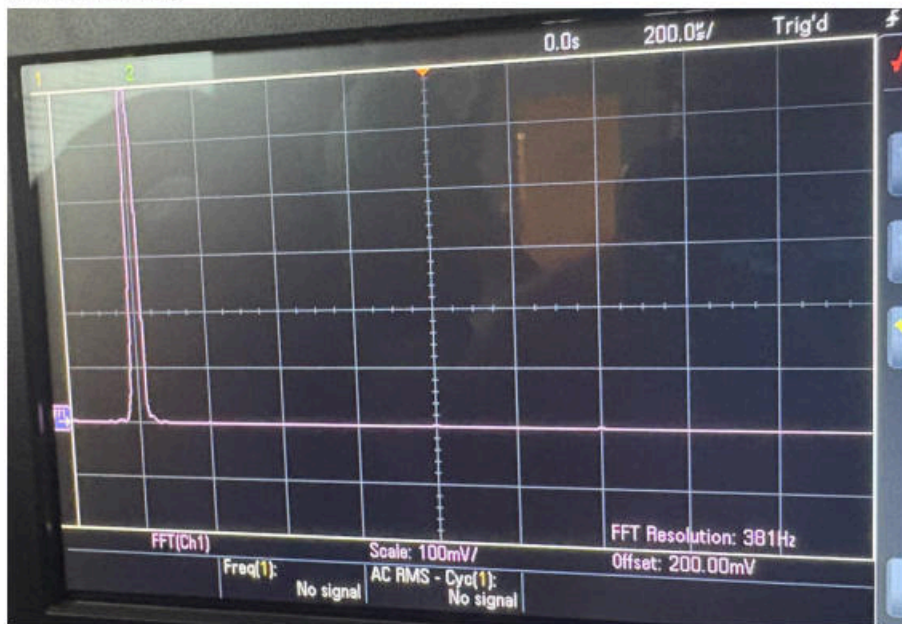


IMG 11. Results



IMG 12. Results

Vrms display



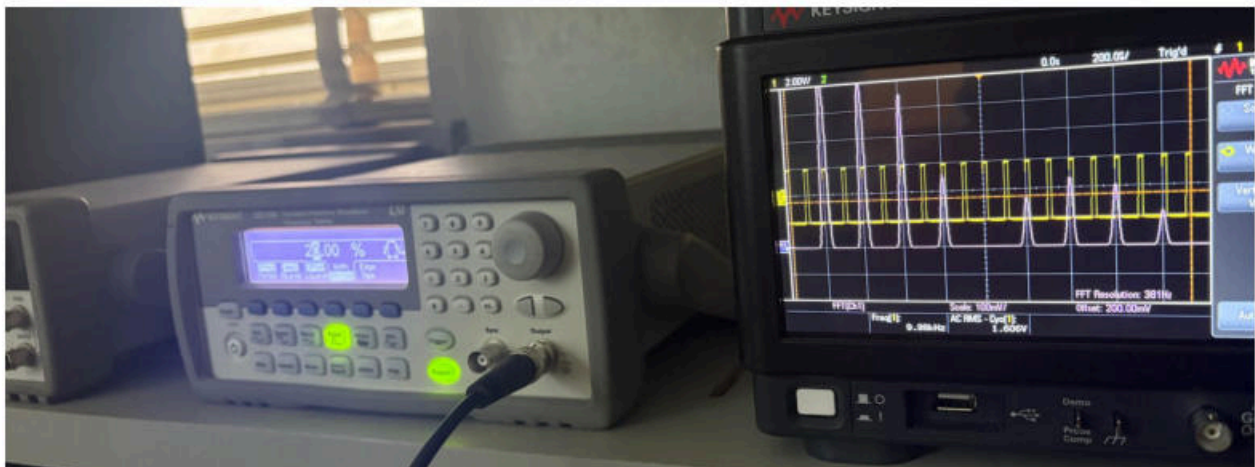
IMG 13. Results

PART 2:

After you feel confident from the FFT tutorial record the following from both the scope function and the FFT, these are the same screens but you might want to only have one displayed at a time to make it clearer. Using the FFT analyzer examine the harmonic content of pulse waveforms and further understand the sinc x, i.e. $\sin x/x$, “envelope” or pattern associated with the harmonics. Use a signal frequency, see tutorial, to display the harmonics such that they are easy to identify.

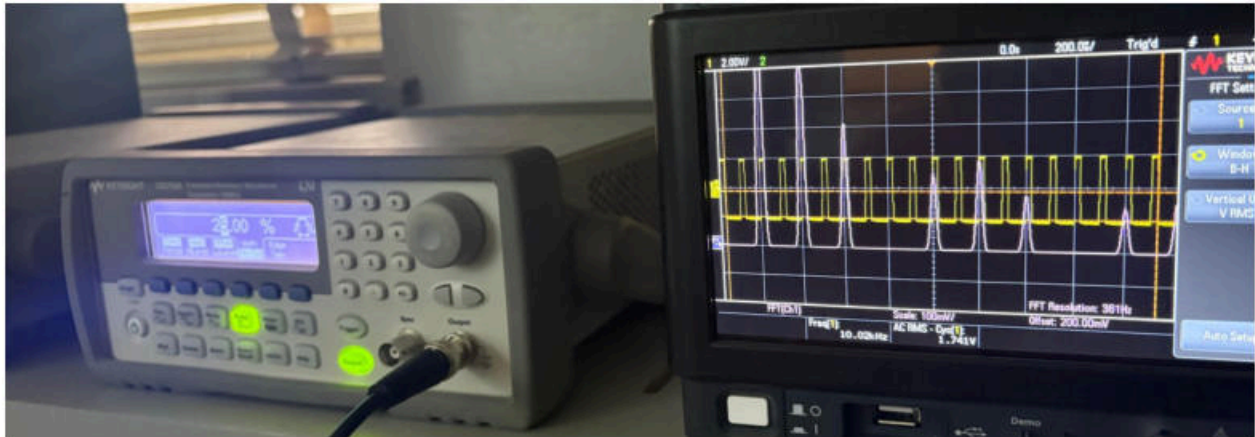
- a) 20% duty cycle signal ($\tau/T = 1/5$), every 5th harmonic zero
- b) 25% duty cycle signal ($\tau/T = 1/4$), every 4th harmonic zero
- c) 33% duty cycle signal ($\tau/T = 1/3$), every 3rd harmonic zero
- d) 50% duty cycle signal ($\tau/T = 1/2$), every 2nd or even harmonic zero

a) 20% duty cycle signal ($\tau/T = 1/5$), every 5th harmonic zero



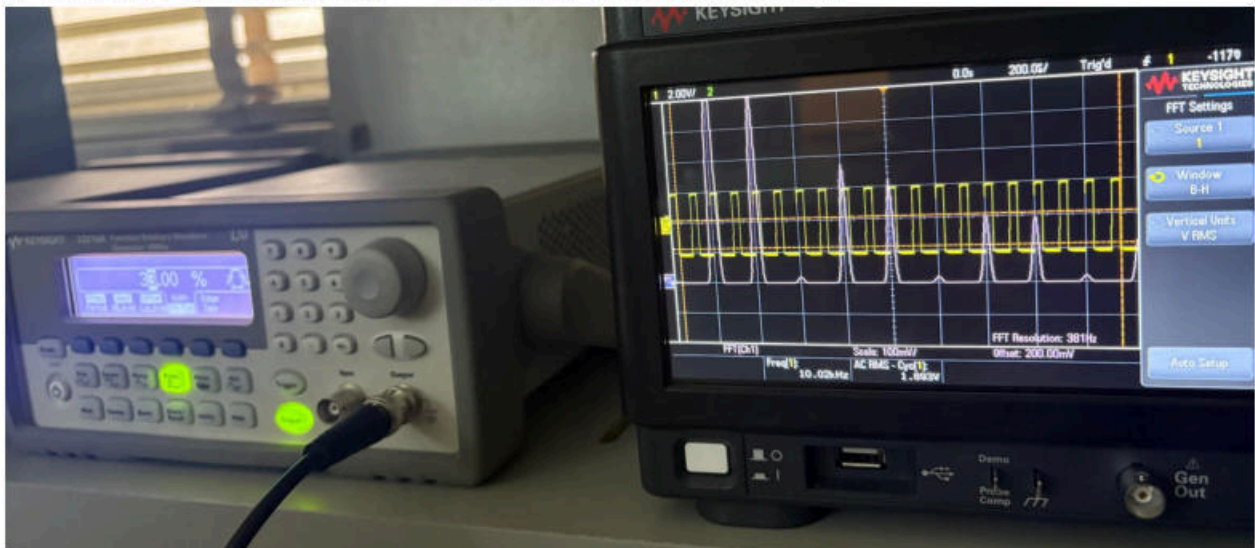
IMG 14. Results

b) 25% duty cycle signal ($\tau/T = 1/4$), every 4th harmonic zero



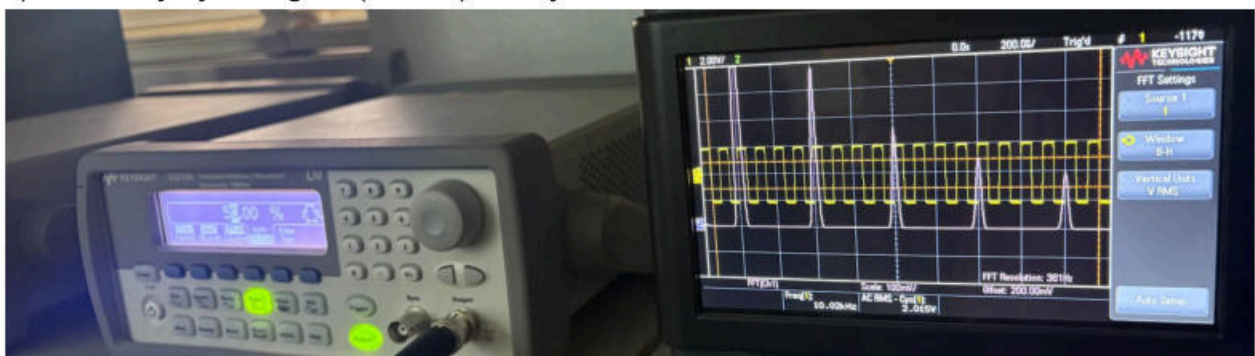
IMG 15. Results

c) 33% duty cycle signal ($T = 1/3$), every 3rd harmonic zero



IMG 16. Results

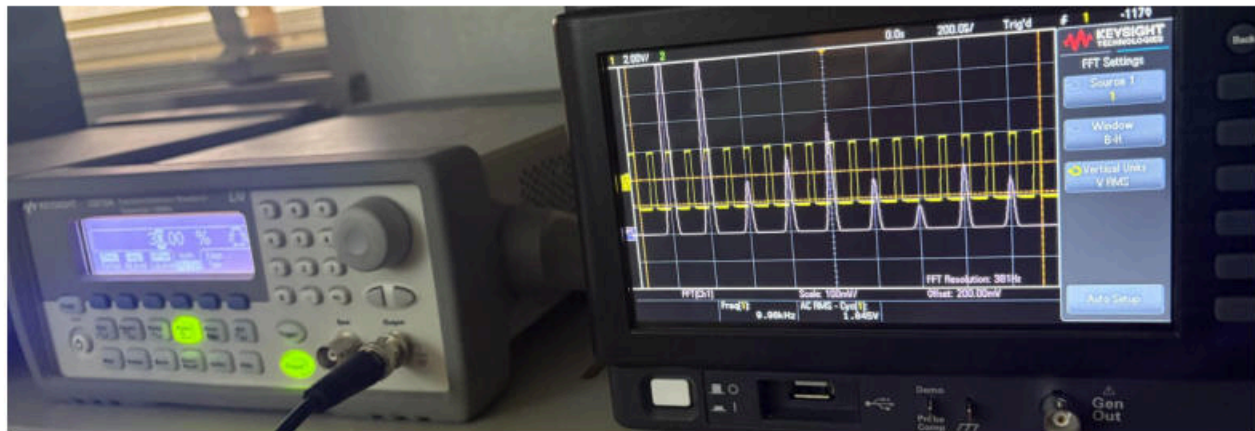
d) 50% duty cycle signal ($T = 1/2$), every 2nd or even harmonic zero



IMG 17. Results

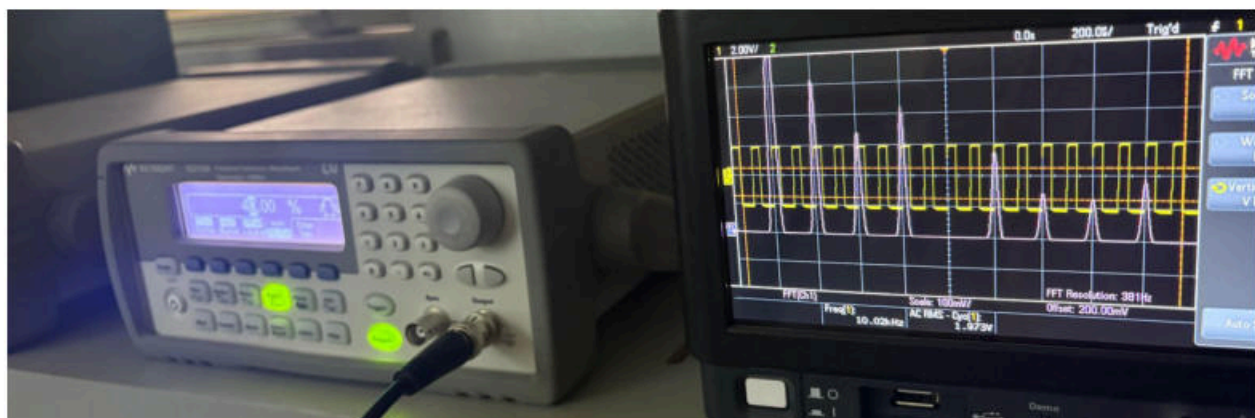
Note: that because the FFT calculation, like SA operation, performs a power-based amplitude calculation the Sinc envelope is an absolute value, i.e. no negative magnitude values are shown for any frequency terms, fundamental or harmonic. After these waveforms examine pulse waveforms with duty cycles of the form m/n with m and n integers and one of them prime.

e) 30% duty cycle signal ($T = 3/10$), every 10th harmonic zero no $10/3$ no $3.333_$ harmonic



IMG 18. Results

f) 40% duty cycle signal ($T = 4/10 = 2/5$), every 5th harmonic zero, no $5/2$ or 2.5 harmonic



IMG 19. Results

PART 3:

3. Change the signal on the ARB to a Sinc signal by selecting the Arb button, below Noise, then

use the right (and left) arrow keys at the bottom of the unit just below and to the right of the display, don't use the larger back arrow keys to the right of the display. After pushing the right arrow key will see the indicators under the display change to Select Wform, Create New etc. select the first one Select Wform then Builtin then Sinc. Capture the displays, time and frequency domain as shown below. Note the "square" response in the frequency domain, fundamental and harmonics nearly the same level, then nearly nothing which points to the duality or interrelationship of the time and frequency domains. As shown this Sinc signal is the time domain whereas previously the pulse or square wave in the time domain resulted in a sinc function in the frequency domain



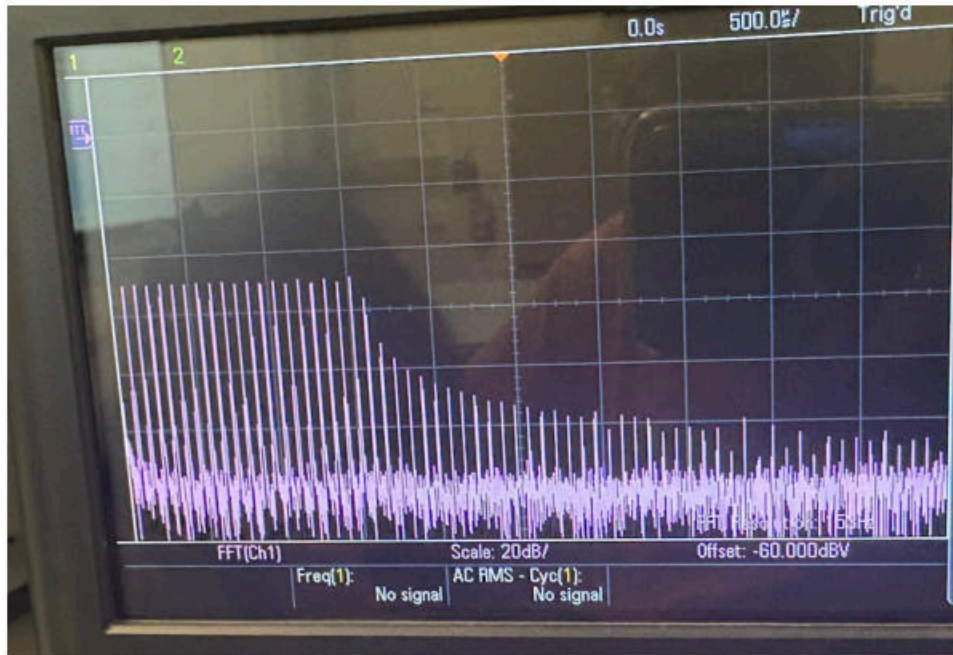
IMG 20.

PART 4:

4. Change the ARB signal to a Cardiac signal (ARB button, right arrow, Select Wform, Builtin Cardiac), again record the time and frequency domain. Note this cardiac is not your heart beat signal as your heart only beats on average once a second (60-100 times per minute, not in kHz) but the frequency content (fundamental and harmonics) would be the same if scaled downward for a fundamental frequency of 1Hz for biomedical work. Note: On your testbench if you are using the Keysight 33210A instead

of the Rigol it may not have the ARB signals enabled in that case use the Agilent 33120A or the Rigol DG5071 unit.

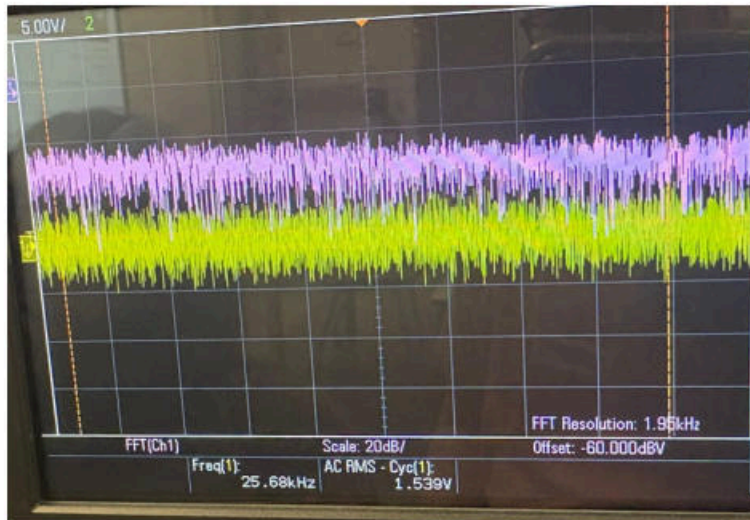
Picture of the cardiac



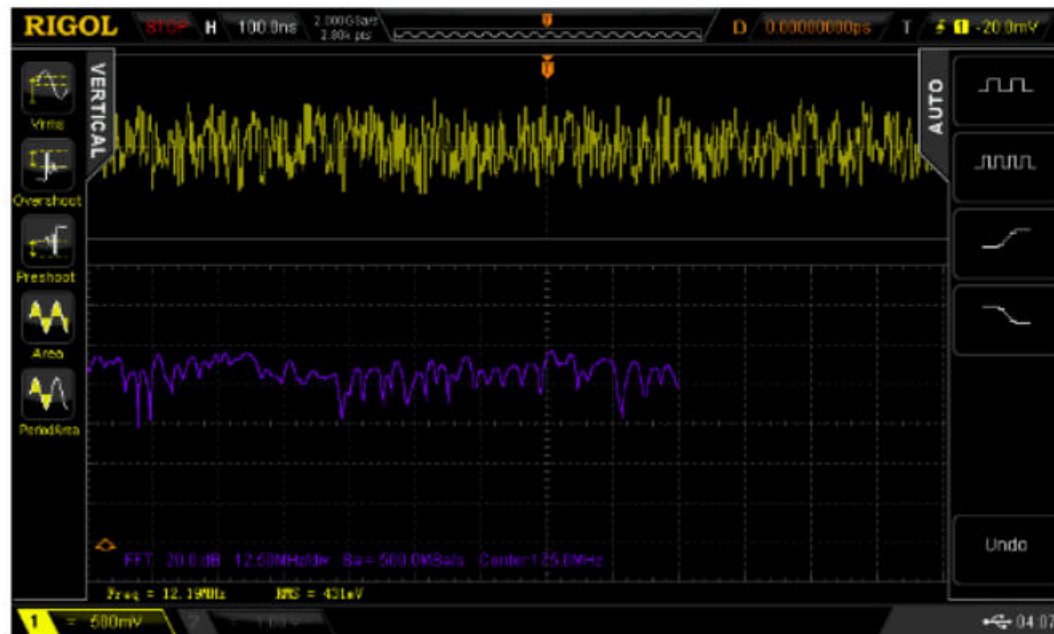
IMG 21. Results

PART 5:

5. Because the above signals (pulse, Sinc and Cardiac have DC or average values it is not recommended to view them on the Spectrum Analyzer which is an RF device but the Noise signal (which is the button above ARB) has 0V DC so it is ok to view it on the Spectrum Analyzer. Select Noise and view and record the time and frequency content with both the SA and scope FFT function. Recommended settings on the SA are Frequency 0Hz to 100MHz, on the scope use auto-scale then SINGLE to display the random noise. Random noise is flat across all frequencies. Note the scope frequency display stops a few divisions above the center of 125MHz



IMG 22. Results



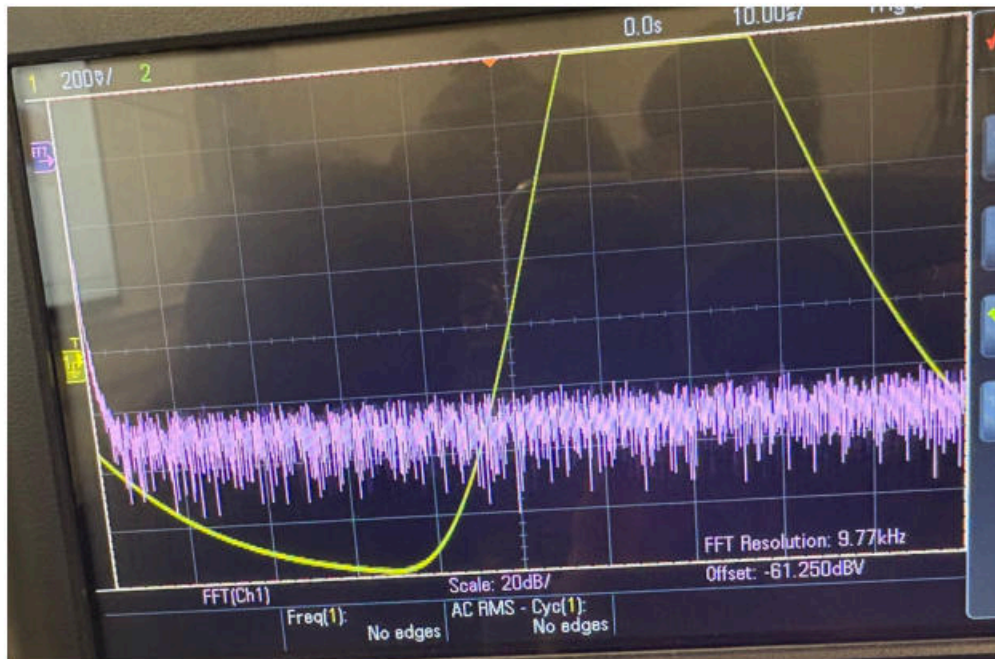
IMG 23. Expected

PART 6: Filtering effects

Filtering Effects: Lastly we will look how filtering affects the Fourier series components and time domain signal with just the signal generator and scope. The farthest right piece of equipment on the bench is the Krohn-Hite active filter, as an active filter it has amplifiers so first off it must be turned on. This unit is a tunable High-Pass Filter (left side controls) in series with a Low-Pass Filter (right side controls) with the HP and LP critical frequencies adjustable from 10Hz to 1MHz. The HP controls on the left side will be set to a minimum or 10Hz to let all signals their harmonics pass while the LP or right side controls will be adjusted to reduce higher harmonic values.

With respect to filtering any waveform could be examined and you are free to choose your own but since square or pulse waves contain many harmonics they are useful to examine. As an example below is a 10kHz 20% pulse wave, in the top left display the LP critical frequency is set at the maximum or >1MHz and as shown there is minimal effect as the output of the filter (blue signal) looks essentially like the input (yellow signal) because all harmonics of this 10kHz signal, up to 1MHz, are allowed to pass through. As the LP frequency (top right display) is reduced to 100 kHz some rounding of the “edges” is noticeable along with some “peaking” as high frequency harmonics are reduced.

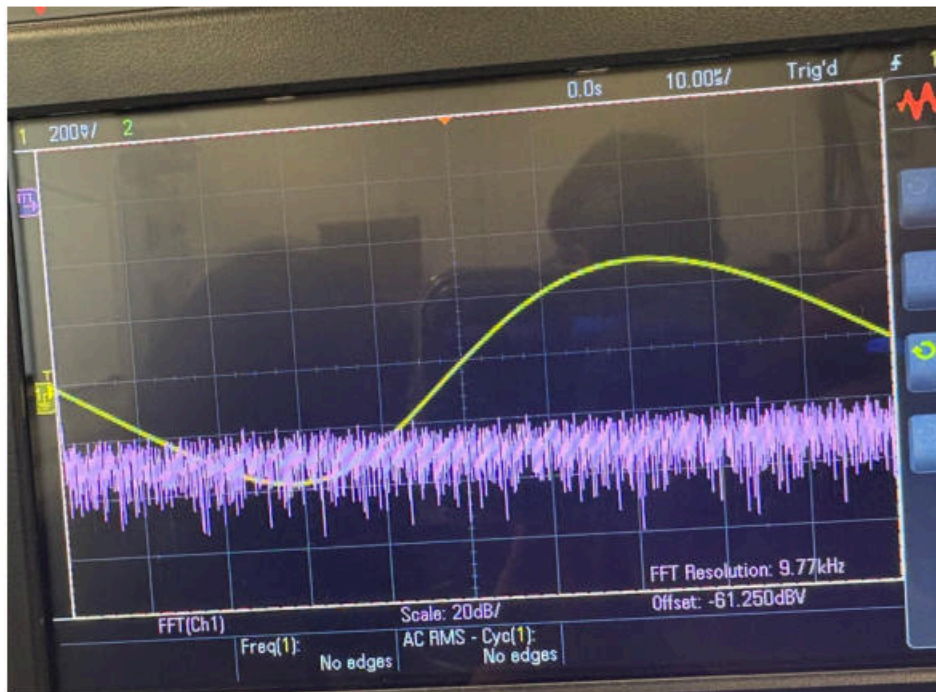
Rounding the edges by adjusting:



IMG 24.



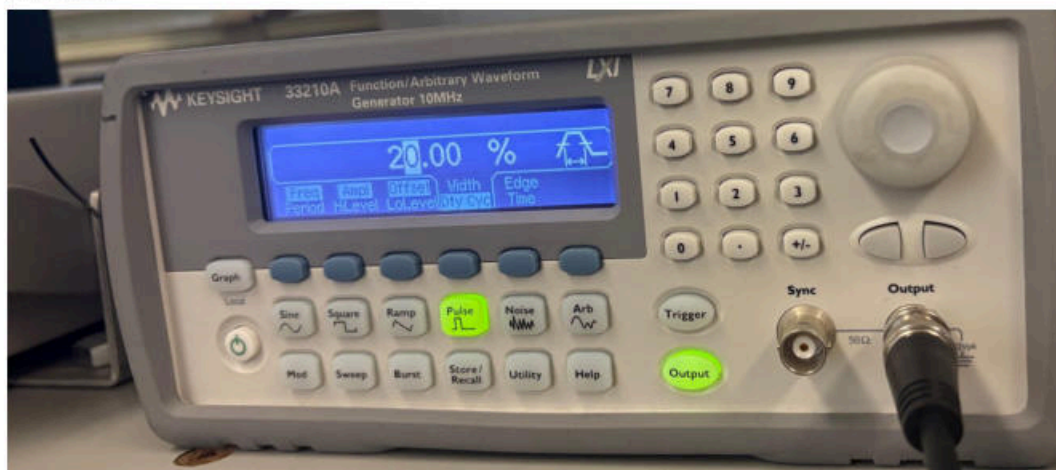
IMG 25.



IMG 26.



IMG 27.

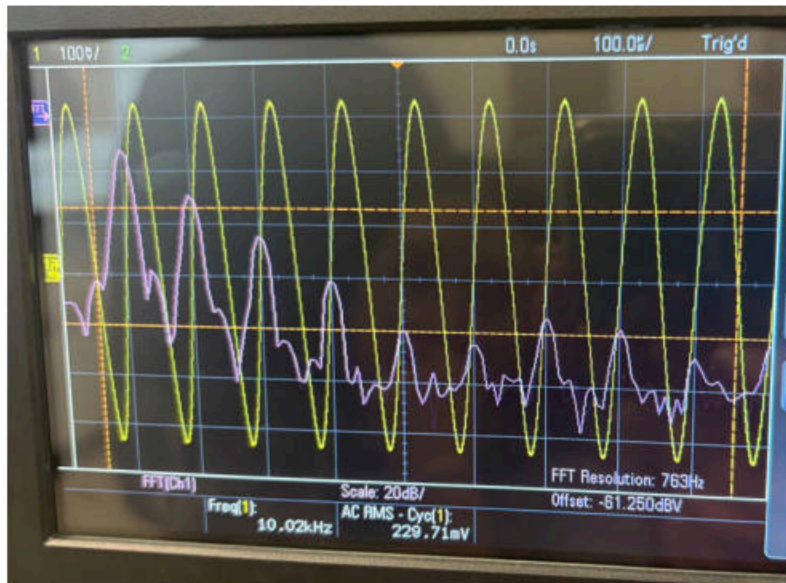


IMG 28.

In the 3rd lower left display the frequency has been reduced to ~42 kHz which only allows the fundamental and few harmonics to pass and so a noticeably rounded and time delayed signal is the effect. Finally with the filter frequency set to around 10 kHz only the time delay and reduced amplitude fundamental sinewave makes it through the filter. Note in the above case the filter was changed but one could see a similar effect if the filter was held constant and the frequency of the input signal changed, try it out and see the effect.



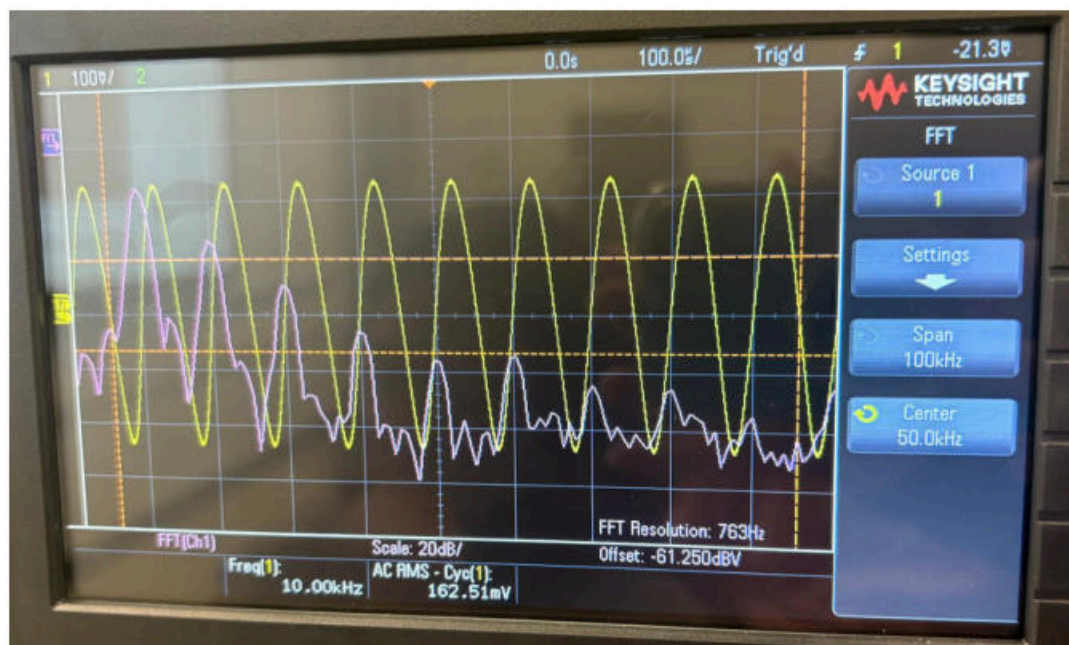
IMG 29.



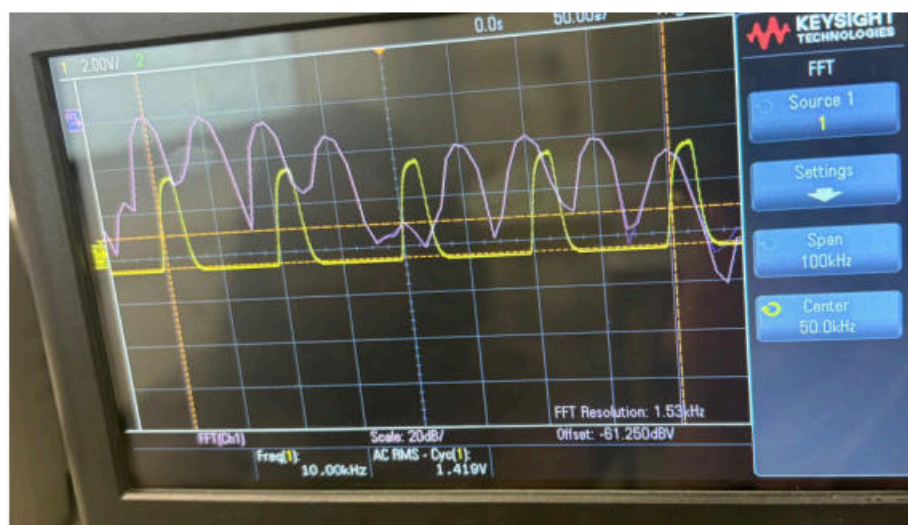
IMG 30.



IMG 31.



IMG 32.



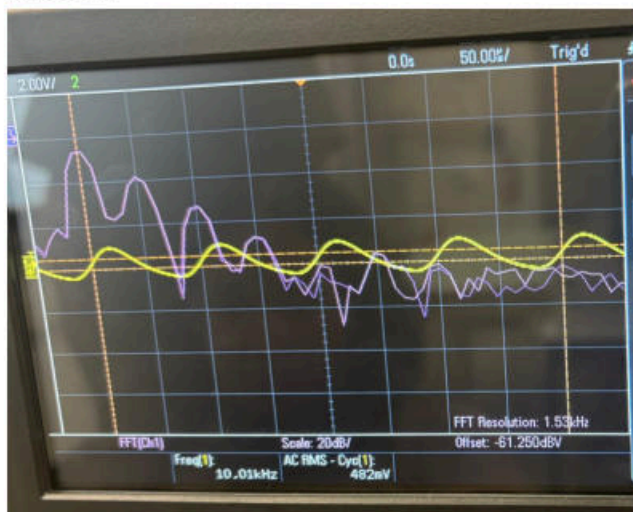
IMG 33.



IMG 34.



IMG 35.



IMG 36.

extra credit

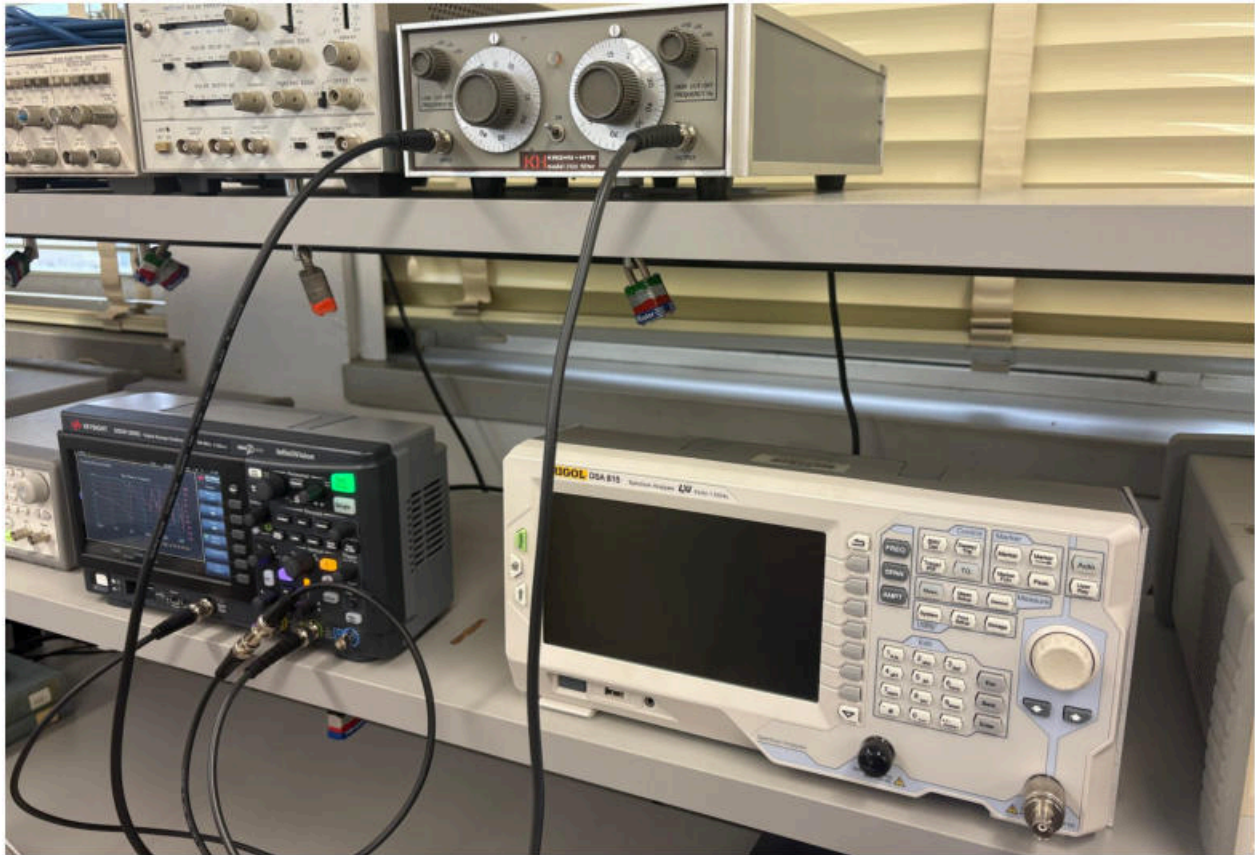
For extra credit take one of the filter cases, middle cases 2 or 3 would be best, and create the Bode plot Gain and Phase of the filter using the new Scope FRA Analysis we learned about earlier in the term. The Bode plot will allow you to determine each harmonic value in magnitude and phase which can then be added up to create an approximation of the output wave.



IMG 37. Blue = LP filter, red = phase



IMG 38.

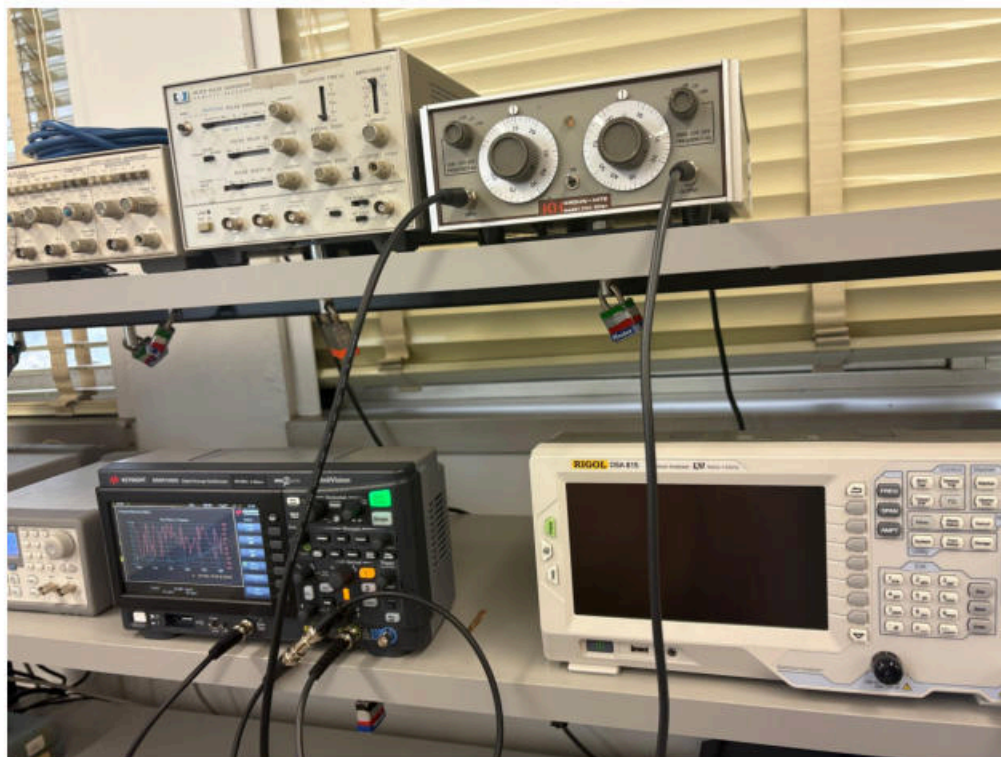


IMG 39.

Heart rate
Cardiac



IMG 40.



IMG 41.



IMG 42.

Conclusion

In this lab the FFT (Fast Fourier Transform) feature of the oscilloscope was used to understand what frequencies make up different electrical signals. Normally signals are viewed in the time domain as voltage changing over time but the FFT allows the same signal to be viewed in the frequency domain (which shows how strong each frequency component of the signal is). The FFT was first tested using a sine wave, which correctly showed a single frequency peak and confirmed that the oscilloscope sampling and FFT settings were working properly.

When square and pulse waveforms were examined, we observed multiple frequency components (called harmonics) at whole number multiples of the main signal frequency. Changing the duty cycle of the pulse waveform caused certain harmonics to disappear which matched the expected behavior seen in the FFT display. The overall shape of the harmonic amplitudes followed a sinc-like pattern, showing that shorter pulses spread energy across more frequencies. Additional waveforms (such as sinc and cardiac signals) were also analyzed and their FFT results included in this lab report showed that these such complex time domain shapes contain many frequency components. Random noise was observed to have a flat frequency spectrum (meaning it contains roughly equal energy at all frequencies., according to the lab handout).

Finally, the effect of filtering was demonstrated by using a low-pass filter to remove high-frequency components from a pulse waveform. As the filter cutoff frequency was lowered the wave became smoother and more rounded as harmonics were removed, eventually resembling a sine wave (no harmonics). This showed that sharp edges in a signal require high-frequency harmonic and removing those harmonics changes the signal's shape. Overall this lab demonstrated that the FFT is a powerful and useful tool for connecting what is seen in the time domain with the frequency content of a signal, showing us why different waveforms look the way they do.